Plasma and urinary alkylresorcinol metabolites as potential biomarkers of breast cancer risk in Finnish women: a pilot study.

Alkylresorcinols (ARs) are shown to be good biomarkers of consumption of rye and whole-grain wheat products in man. The aim of this pilot study was to investigate AR metabolites as potential biomarkers of breast cancer (BC) risk in Finnish women since intake of cereal fiber and its components has been proposed to reduce this risk through an effect on the enterohepatic circulation of estrogens. This was a cross-sectional and observational pilot study. A total of 20 omnivores, 20 vegetarians, and 16 BC women (6-12 mo after operation) were investigated on 2 occasions 6 mo apart. Dietary intake (5-days record), plasma/urinary AR metabolites [3,5-dihydroxybenzoic acid (DHBA) and 3-(3,5-dihydroxyphenyl)-1-propanoic acid (DHPPA)] and plasma/urinary enterolactone were measured. The groups were compared using nonparametric tests. We observed that plasma DHBA ($P = 0.007$; $P = 0.03$), plasma DHPPA ($P = 0.02$; $P = 0.01$), urinary DHBA ($P = 0.001$; $P = 0.003$), urinary DHPPA ($P = 0.001$; $P = 0.001$), and cereal fiber intake ($P = 0.007$; $P = 0.003$) were significantly lower in the BC group compared to the vegetarian and omnivore groups, respectively. Based on measurements of AR metabolites in urine and in plasma, whole-grain rye and wheat cereal fiber intake is low in BC subjects. Thus, urinary and plasma AR metabolites may be used as potential biomarkers of BC risk in women. This novel approach will likely also facilitate studies of associations between rye and whole-grain wheat cereal fiber intake and other diseases. Our findings should, however, be confirmed with larger subject populations.